

Interpretable Multi-Modal Reasoning Framework for Protein Stability Prediction: Bridging Deep Learning and Rational Design

Abigail Lin*

Department of Computer Science, University of Florida

Abstract

Predicting the effects of amino acid mutations on protein stability ($\Delta\Delta G$) is crucial for protein engineering, drug design, and understanding disease mechanisms. While recent deep learning approaches have achieved impressive predictive accuracy, they often lack interpretability and strategic reasoning capabilities essential for practical protein design tasks. We present a novel multi-modal framework that combines Graph Neural Networks (GNNs) for 3D structural information with Transformers for sequence modeling, augmented with multiple interpretable reasoning strategies for mutation selection. Our framework introduces: (1) a cross-modal fusion architecture that integrates spatial and sequential protein features, (2) mutation-aware attention mechanisms that focus on local structural perturbations, (3) five distinct search strategies (greedy, beam search, Monte Carlo, ensemble, and constraint-based) for exploring mutation space, and (4) chain-of-thought reasoning modules that generate human-interpretable explanations for each prediction. Evaluated on benchmark datasets including ProTherm and SKEMPI, our approach achieves competitive accuracy (Pearson $\rho = 0.85$) while providing actionable insights through molecular rationale generation. The framework successfully identifies stabilizing mutations with confidence scores and contextual awareness, demonstrated through case studies in antibody engineering and enzyme thermostability optimization. This work bridges the gap between predictive models and practical protein engineering by providing both accurate predictions and interpretable decision-making processes.

Keywords: Protein engineering, $\Delta\Delta G$ prediction, Graph neural networks, Transformers, Multi-modal learning, Interpretable AI, Rational protein design

1 Introduction

1.1 Background and Motivation

Protein thermodynamic stability, quantified by the change in Gibbs free energy upon mutation ($\Delta\Delta G = G_{\text{mutant}} - G_{\text{wild-type}}$), is a fundamental property that governs protein folding, function, and evolutionary fitness [Tokuriki and Tawfik, 2009]. Accurate prediction of mutation effects enables rational protein engineering for diverse applications including therapeutic antibody development [?], industrial enzyme optimization [?], and understanding disease-causing mutations [?].

Traditional computational methods employ physics-based energy functions (FoldX [Schymkowitz et al., 2005b], Rosetta [Kellogg et al., 2011b]), achieving moderate accuracy (Pearson $\rho \approx 0.5$) but requiring extensive computational resources. Recent machine learning approaches leverage deep neural networks, improving accuracy to $\rho \approx 0.75 - 0.80$ [Meier et al., 2021a, Cao et al., 2019]. However, these methods face three critical limitations:

*Corresponding author: abigail.lin@ufl.edu

Challenge 1: Interpretability Gap. Black-box models provide predictions without mechanistic explanations, hindering adoption by experimental researchers who need to understand *why* a mutation is predicted to stabilize or destabilize a protein.

Challenge 2: Modal Integration. Proteins exhibit multi-scale organization—sequence determines structure, which determines function. Existing methods process sequence and structure independently, missing critical cross-modal dependencies.

Challenge 3: Strategic Reasoning. Protein design requires systematic exploration of mutation space with multiple competing objectives (stability, solubility, activity). Current approaches evaluate mutations individually without strategic search capabilities.

1.2 Our Contributions

We present an **Interpretable Multi-Modal Reasoning Framework** that addresses these challenges through:

1. **Hybrid GNN-Transformer Architecture (§??):** Novel cross-modal fusion integrating geometric graph neural networks for 3D structure with transformers for sequence, using mutation-aware attention mechanisms.
2. **Strategic Search Framework (§??):** Five complementary strategies (greedy, beam search, Monte Carlo, ensemble, constraint-based) with rigorous mathematical formulation and convergence analysis.
3. **Chain-of-Thought Reasoning (§??):** Automated generation of human-interpretable explanations grounded in structural biology principles, with confidence calibration and molecular rationale.
4. **Comprehensive Evaluation (§??):** Achieves state-of-the-art performance ($\rho = 0.85$) on ProTherm with superior interpretability (human evaluation: 4.2/5.0), demonstrated through three real-world case studies.

1.3 Paper Organization

Section 2 reviews related work. Section 3 details the prediction model architecture. Section 4 presents the reasoning framework with mathematical formulation. Section 5 describes experiments and results. Section 6 discusses findings and limitations. Section 7 concludes with future directions.

2 Related Work

2.1 $\Delta\Delta G$ Prediction Methods

Physics-based approaches [Schymkowitz et al., 2005b, Kellogg et al., 2011b] use molecular mechanics force fields with energy minimization. While interpretable, they achieve limited accuracy and are computationally expensive (minutes per mutation).

Statistical methods [??] use hand-crafted features with classical ML (SVM, random forests), achieving $\rho \approx 0.6 - 0.7$.

Sequence-based deep learning [Meier et al., 2021a, Rives et al., 2021] employs protein language models pre-trained on millions of sequences, reaching $\rho \approx 0.75 - 0.78$ but ignoring 3D structure.

Structure-based deep learning [Cao et al., 2019, Li et al., 2020a] uses 3D CNNs or GNNs, achieving $\rho \approx 0.70 - 0.73$ but lacking sequence evolutionary context.

Hybrid approaches [Zhou et al., 2020] combine both modalities via simple concatenation or late fusion, reaching $\rho \approx 0.76$ but without principled cross-modal interaction.

Gap: None provide interpretable reasoning or strategic mutation search capabilities.

2.2 Interpretable AI for Biology

Recent work explores attention visualization [?], saliency maps [?], and concept-based explanations [?]. However, these provide post-hoc explanations rather than integrated reasoning. Our chain-of-thought approach generates explanations as part of the prediction process, grounded in domain knowledge.

2.3 Protein Design and Optimization

Traditional design uses iterative rounds of prediction and experimental testing. Recent computational methods employ: - Genetic algorithms [?] - Bayesian optimization [?] - Reinforcement learning [?]

Gap: These methods lack interpretability and don’t provide mechanistic reasoning for design choices. Our framework integrates multiple search strategies with explicit reasoning chains.

3 Methodology

3.1 Problem Formulation

Input:

- Protein sequence $\mathbf{s} = (s_1, \dots, s_L)$ where $s_i \in \mathcal{A}$ (20 amino acids)
- 3D structure graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V}$ represents residues with coordinates $\{\mathbf{r}_i\}$
- Mutation $m = (p, s_p^{\text{wt}}, s_p^{\text{mut}})$: position, wild-type, mutant

Output:

- $\Delta\Delta G$ prediction with confidence $c \in [0, 1]$
- Reasoning chain $\mathcal{R} = \{r_1, \dots, r_k\}$ explaining the prediction
- Molecular rationale connecting prediction to structural/sequence features

References

Habib Bashour, Eva Smorodina, Matteo Pariset, Jahn Zhong, Rahmad Akbar, Maria Chernigovskaya, Khang Lê Quý, Igor Snapkow, Puneet Rawat, Konrad Krawczyk, Geir Kjetil Sandve, Jose Gutierrez-Marcos, Daniel Nakhaee-Zadeh Gutierrez, Jan Terje Andersen, and Victor Greiff. Biophysical cartography of the native and human-engineered antibody landscapes quantifies the plasticity of antibody developability. *Communications Biology*, 7(1):922, Jul 2024. ISSN 2399-3642. doi: 10.1038/s42003-024-06561-3. URL <https://doi.org/10.1038/s42003-024-06561-3>.

- Markus Braun, Adrian Tripp, Morakot Chakatok, Sigrid Kaltenbrunner, Massimo Totaro, David Stoll, Aleksandar Bijelic, Wael Elaily, Shlomo Yakir Hoch, Matteo Aleotti, Mélanie Hall, and Gustav Oberdorfer. Computational design of highly active de novo enzymes. *bioRxiv*, 2024. doi: 10.1101/2024.08.02.606416. URL <https://www.biorxiv.org/content/early/2024/08/03/2024.08.02.606416>.
- Markus Braun, Adrian Tripp, Morakot Chakatok, Sigrid Kaltenbrunner, Celina Fischer, David Stoll, Aleksandar Bijelic, Wael Elaily, Massimo G. Totaro, Melanie Moser, Shlomo Y. Hoch, Horst Lechner, Federico Rossi, Matteo Aleotti, Mélanie Hall, and Gustav Oberdorfer. Computational enzyme design by catalytic motif scaffolding. *bioRxiv*, 2025. doi: 10.1101/2024.08.02.606416. URL <https://www.biorxiv.org/content/early/2025/06/11/2024.08.02.606416>.
- Hong Cao, Jielin Wang, Lin He, Yan Chen, Kaixian Chen, Hualiang Jiang, and Jian Li. Deepddg: Predicting the stability change of protein point mutations using neural networks. *Journal of Chemical Information and Modeling*, 59(4):1508–1514, 2019.
- Tanya Jain, Tong Sun, Stéphane Durand, Andrew Hall, Nathan R. Houston, Jason H. Nett, Brian Sharkey, Boguslaw Bobrowicz, Isaac Caffry, Yingda Yu, Yi Cao, Heather Lynaugh, Michael Brown, Himadri Baruah, Laura T. Gray, Eric M. Krauland, Yan Xu, Martha Vásquez, and K. Dane Wittrup. Biophysical properties of the clinical-stage antibody landscape. *Proceedings of the National Academy of Sciences of the United States of America*, 114(5):944–949, January 2017. doi: 10.1073/pnas.1616408114. URL <https://doi.org/10.1073/pnas.1616408114>.
- Jovita Jankauskaitė, Bruno Jiménez-García, Justas Dapkūnas, Joost Schymkowitz, Ignacio Ferreira, Francesco Luigi Gervasio, Johannes Sódíng, Wim Vranken, Adrián Velazquez-Campoy, and Yves Dehouck. Skempi 2.0: an updated benchmark of changes in protein–protein binding energy, kinetics and thermodynamics upon mutation. *Bioinformatics*, 35(3):462–469, 2019.
- Bowen Jing, Stephan Eismann, Patricia A. Suriana, Raphael J. L. Townshend, and Ron Dror. Learning from protein structure with geometric vector perceptrons. In *International Conference on Learning Representations (ICLR)*, 2021.
- Wolfgang Kabsch and Chris Sander. Dictionary of protein secondary structure: pattern recognition of hydrogen-bonded and geometrical features. *Biopolymers*, 22(12):2577–2637, 1983.
- Elizabeth H. Kellogg, Andrew Leaver-Fay, and David Baker. Role of conformational sampling in computing mutation-induced changes in protein structure and stability. *Proteins: Structure, Function, and Bioinformatics*, 79(3):830–838, March 2011a. doi: 10.1002/prot.22921. URL <https://doi.org/10.1002/prot.22921>.
- Elizabeth H Kellogg, Andrew Leaver-Fay, and David Baker. Role of conformational sampling in computing mutation-induced changes in protein structure and stability. *Proteins: Structure, Function, and Bioinformatics*, 79(3):830–838, 2011b.
- Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations (ICLR)*, 2017.
- Johannes Klicpera, Stefan Weißenberger, and Stephan Günnemann. Diffusion improves graph learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, pages 13340–13352, 2019.

- M D Kumar, K A Bava, M Michael Gromiha, N Srinivasan Prabhu, Shandar Ahmad, Harinder Singh, and Akinori Sarai. Protherm and pronit: thermodynamic databases for proteins and protein–nucleic acid interactions. *Nucleic Acids Research*, 34(D1):D204–D206, 2006.
- Bo Li, Yung-Tsung Yang, John A Capra, and Mark B Gerstein. Predicting changes in protein thermodynamic stability upon point mutation with deep 3d convolutional neural networks. *PLoS Computational Biology*, 16(11):e1008291, 2020a.
- Guohao Li, Chen Xiong, Ali Thabet, and Bernard Ghanem. Deepergcn: All you need to train deeper gcn. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2020b.
- Dina Listov, Eva Vos, Gyula Hoffka, Shlomo Yakir Hoch, Andrej Berg, Shelly Hamer-Rogotner, Orly Dym, Shina Caroline Lynn Kamerlin, and Sarel J. Fleishman. Complete computational design of high-efficiency kemp elimination enzymes. *Nature*, 643(8074):1421–1427, Jul 2025. ISSN 1476-4687. doi: 10.1038/s41586-025-09136-2. URL <https://doi.org/10.1038/s41586-025-09136-2>.
- Joshua Meier, Roshan Rao, Raoul Verkuil, Jason Liu, Tom Sercu, Mihaly Varadi, Jinhe Lee, John Muschelli, Dima Kozakov, Andrej Sali, et al. Language models enable zero-shot prediction of the effects of mutations on protein function. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 29287–29303, 2021a.
- Joshua Meier, Roshan Rao, Robert Verkuil, Jason Liu, Tom Sercu, and Alex Rives. Language models enable zero-shot prediction of the effects of mutations on protein function. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 29287–29303. Curran Associates, Inc., 2021b. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/f51338d736f95dd42427296047067694-Paper.pdf.
- Luca Montanucci, Emidio Capriotti, Yana Frank, Pier Luigi Martelli, and Rita Casadio. DDGun: an untrained predictor of protein stability changes upon amino acid variants. *Nucleic Acids Research*, 50(W1):W222–W227, 2022.
- Roshan Rao, Joshua Meier, Tom Sercu, Saleh Manzoor, Wojciech Martini, Aditya Benites, Calvin Pan, Jesse Ingraham, Stephane Gobeil, Fatima Amanat, et al. Transformer protein language models are unsupervised structure learners. In *International Conference on Learning Representations (ICLR)*, 2021.
- Alexander Rives, Joshua Meier, Tom Sercu, Siddharth Goyal, Zeming Lin, Jason Guo, Myle Ott, C Lawrence Zitnick, Jerry Ma, and Rob Fergus. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences (PNAS)*, 118(15):e2016239118, 2021.
- Joost Schymkowitz, Jan Borg, Florian Stricher, Robin Nys, Frederic Rousseau, and Luis Serrano. The foldx web server: an online force field. *Nucleic Acids Research*, 33(Web Server issue):W382–W388, July 2005a. doi: 10.1093/nar/gki387. URL <https://doi.org/10.1093/nar/gki387>.
- Joost Schymkowitz, Jesper Borg, Francois Stricher, Remy Nys, Frederic Rousseau, and Luis Serrano. The FoldX web server: an online force field. *Nucleic Acids Research*, 33(suppl_1):W382–W388, 2005b.

- Jan Stourac, Jiri Dubrava, Milos Musil, David Bednar, Jiri Damborsky, and Zbynek Prokop. Fire-protdb: database of manually curated protein stability data. *Nucleic Acids Research*, 49(D1): D319–D324, 2021.
- Nobuhiko Tokuriki and Dan S Tawfik. Stability effects of mutations and protein evolvability. *Current Opinion in Structural Biology*, 19(5):596–604, 2009. doi: 10.1016/j.sbi.2009.08.003.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.
- Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In *International Conference on Learning Representations (ICLR)*, 2018.
- Peiqi Yue, Zhen Li, and John Moult. Loss of protein structure stability as a major causative factor in monogenic disease. *Journal of Molecular Biology*, 353(2):459–473, October 2005. doi: 10.1016/j.jmb.2005.08.020.
- Guangyu Zhou, Min Chen, Christopher J T Ju, Jie Xu, Haoyang Chen, and Zhiyong Lu. Mutation effect estimation on protein–protein interactions using deep contextualized representation learning. *NAR Genomics and Bioinformatics*, 2(2):lqaa015, 2020.
- Yunzhuo Zhou, Qisheng Pan, Douglas E V Pires, Carlos H M Rodrigues, and David B Ascher. Ddmut: predicting effects of mutations on protein stability using deep learning. *Nucleic Acids Research*, 51(W1):W122–W128, 06 2023. ISSN 0305-1048. doi: 10.1093/nar/gkad472. URL <https://doi.org/10.1093/nar/gkad472>.